
100 JavaScript Fundamentals MCQs

Part 1: History, Standards & Evolution

1. **Who created JavaScript?**
 - A) Brendan Eich
 - B) James Gosling
 - C) Tim Berners-Lee
 - D) Guido van Rossum
2. **In which year was JavaScript first released?**
 - A) 1991
 - B) 1995
 - C) 1998
 - D) 2000
3. **What is the official name of the standard upon which JavaScript is based?**
 - A) JavaScript Standard
 - B) ECMA-262
 - C) ISO-JS
 - D) W3C Script
4. **Which version of ECMAScript introduced major modern features like `let`, `const`, and Arrow Functions?**
 - A) ES3
 - B) ES5
 - C) ES6 (ES2015)
 - D) ES2020
5. **JavaScript was originally developed at which company?**
 - A) Microsoft
 - B) Sun Microsystems
 - C) Netscape
 - D) Oracle

Part 2: Variables (`var`, `let`, `const`)

6. **Which keyword declares a variable that can be redeclared in the same scope?**
 - A) `let`
 - B) `const`
 - C) `var`
 - D) `static`
7. **What is the scope of a variable declared with `let`?**
 - A) Global scope only
 - B) Function scope

- C) Block scope
 - D) Module scope
8. **Which statement is true regarding `const`?**
- A) It makes the variable value immutable (for primitives).
 - B) It prevents the object properties from being changed.
 - C) It allows reassignment.
 - D) It is hoisted and initialized with `undefined`.
9. **What happens if you try to log a `let` variable before it is declared?**
- A) It prints `undefined`.
 - B) It prints `null`.
 - C) It throws a `ReferenceError`.
 - D) It throws a `SyntaxError`.
10. **Which of the following creates a constant reference to an object?**
- A) `var obj = {}`
 - B) `let obj = {}`
 - C) `const obj = {}`
 - D) `fixed obj = {}`
11. **Variables declared with `var` are attached to which object in the browser global scope?**
- A) `document`
 - B) `console`
 - C) `window`
 - D) `navigator`
12. **Can you update the properties of an object declared with `const`?**
- A) Yes
 - B) No
 - C) Only if strict mode is off
 - D) Only inside a function
13. **What is the "Temporal Dead Zone"?**
- A) The time before the page loads.
 - B) The period between entering a scope and the actual declaration of a `let/const` variable.
 - C) A deprecated feature of ES5.
 - D) The gap between client and server request.
14. **Which variable declaration is hoisted but NOT initialized?**
- A) `var`
 - B) `let`
 - C) `function`
 - D) `global`
15. **Which is the best practice for declaring variables in modern JavaScript?**
- A) Always use `var`.
 - B) Use `let` for everything.
 - C) Use `const` by default, `let` if reassignment is needed.
 - D) Do not use keywords.

Part 3: Data Types & Primitives

16. Which of the following is NOT a primitive data type in JavaScript?
 - A) Boolean
 - B) Undefined
 - C) Object
 - D) Symbol
17. What is the type of NaN?
 - A) Object
 - B) Number
 - C) String
 - D) Undefined
18. Which data type was added in ES6 to create unique identifiers?
 - A) BigInt
 - B) Unique
 - C) Symbol
 - D) Set
19. What is the result of typeof null?
 - A) "null"
 - B) "undefined"
 - C) "object"
 - D) "number"
20. Which type allows you to safely store integers larger than $2^{53} - 1$?
 - A) LargeInt
 - B) BigNumber
 - C) BigInt
 - D) Double
21. What is the value of a variable that is declared but not assigned?
 - A) null
 - B) 0
 - C) undefined
 - D) false
22. Which of the following is a falsy value?
 - A) "false"
 - B) -1
 - C) "" (Empty String)
 - D) {}
23. Strings in JavaScript are:
 - A) Mutable
 - B) Immutable
 - C) Arrays of characters
 - D) Objects by default
24. How do you denote a BigInt literal?
 - A) 100b

- B) 100n
- C) Big(100)
- D) 100L

25. What is the result of `1 + "2"`?

- A) 3
- B) "12"
- C) NaN
- D) Error

Part 4: Operators & Expressions

26. Which operator checks for both value and type equality?

- A) `==`
- B) `===`
- C) `=`
- D) `!=`

27. What does the ternary operator replace?

- A) for loop
- B) switch statement
- C) if...else statement
- D) while loop

28. What is the result of `!!1`?

- A) true
- B) false
- C) 1
- D) undefined

29. Which operator calculates the remainder of a division?

- A) `/`
- B) `%`
- C) `&`
- D) `rem`

30. What is the result of `false || "Hello"`?

- A) false
- B) true
- C) "Hello"
- D) undefined

31. What is the short-circuit behavior of `&&`?

- A) It stops if the first operand is true.
- B) It stops if the first operand is false.
- C) It always evaluates both operands.
- D) It returns the first operand always.

32. Which operator is used for exponentiation (ES7)?

- A) `^`
- B) `**`

- C) `exp()`
- D) `^^`
- 33. What does `x += 5` mean?
 - A) `x = 5`
 - B) `x = x + 5`
 - C) `x = 5 + 5`
 - D) `x = x * 5`
- 34. Which comparison returns true?
 - A) `5 == "5"`
 - B) `5 === "5"`
 - C) `true == "true"`
 - D) `null === undefined`
- 35. The Nullish Coalescing Operator (`??`) checks for:
 - A) Any falsy value
 - B) Only null or undefined
 - C) Only null
 - D) 0 or false

Part 5: Control Flow (Loops & Conditionals)

- 36. Which loop is guaranteed to execute at least once?
 - A) `for`
 - B) `while`
 - C) `do...while`
 - D) `for...in`
- 37. Which statement stops the current iteration of a loop and continues to the next?
 - A) `stop`
 - B) `break`
 - C) `continue`
 - D) `return`
- 38. The `switch` statement uses which comparison algorithm?
 - A) Loose equality (`==`)
 - B) Strict equality (`===`)
 - C) `Object.is()`
 - D) String conversion
- 39. `for...of` is best used for iterating over:
 - A) Object properties
 - B) Arrays and iterable objects
 - C) Only numbers
 - D) CSS styles
- 40. `for...in` is best used for iterating over:
 - A) Array values
 - B) Object keys (properties)

- C) Map values
- D) Set values

Part 6: Console & Reference Types

41. Which method outputs data as a table in the console?
- A) `console.log()`
 - B) `console.dir()`
 - C) `console.table()`
 - D) `console.grid()`
42. Arrays in JavaScript are of type:
- A) Array
 - B) List
 - C) Object
 - D) Collection
43. How do you access the last element of an array `arr`?
- A) `arr.last()`
 - B) `arr[arr.length - 1]`
 - C) `arr[-1]`
 - D) `arr.end`
44. Which bracket notation is valid for accessing an object property with a space?
- A) `obj.my prop`
 - B) `obj."my prop"`
 - C) `obj["my prop"]`
 - D) `obj(my prop)`
45. If `let a = [1, 2]; let b = a;`, what happens if you do `b.push(3)`?
- A) `a` remains `[1, 2]`.
 - B) `a` becomes `[1, 2, 3]`.
 - C) `b` becomes `undefined`.
 - D) Error occurs.

Part 7: Functions (Basics & Scope)

46. How do you define a default parameter value?
- A) `function myFunc(x = 10) {}`
 - B) `function myFunc(x : 10) {}`
 - C) `function myFunc(x default 10) {}`
 - D) `function myFunc(x) { x = 10 }`
47. Arguments passed to functions are known as:
- A) Parameters
 - B) Props
 - C) Attributes
 - D) Inputs

48. **Primitive types are passed to functions by:**
- A) Reference
 - B) Value
 - C) Pointer
 - D) Inheritance
49. **Objects are passed to functions by:**
- A) Value
 - B) Reference (Reference Copy)
 - C) Cloning
 - D) Stringifying
50. **What is the arguments object in a function?**
- A) An array of all variables in scope.
 - B) An array-like object containing passed arguments.
 - C) A string representation of the function.
 - D) The return value.
51. **A function that calls itself is called:**
- A) Looping
 - B) Recursive
 - C) Circular
 - D) Nested
52. **What keyword is used to return a value from a function?**
- A) send
 - B) output
 - C) return
 - D) yield
53. **Global variables are accessible:**
- A) Only inside functions
 - B) Only inside blocks
 - C) Everywhere in the script
 - D) Only in the file they are defined (modules exception)
54. **What happens if a function does not return a value?**
- A) It returns null.
 - B) It returns 0.
 - C) It returns undefined.
 - D) It throws an error.
55. **Variable length arguments in ES6 are handled using:**
- A) The arguments object
 - B) The spread operator (...) as rest parameters
 - C) varargs
 - D) params[]

Part 8: Advanced Functions (Arrow, Scoping, Closures)

56. Which syntax defines an Arrow Function?

- A) `function() {}`
- B) `() => {}`
- C) `() -> {}`
- D) `=> {}`

57. Arrow functions do NOT have their own:

- A) Return value
- B) Parameters
- C) `this` binding
- D) Scope

58. If an arrow function has a single expression body, what is implied?

- A) It must use curly braces.
- B) Implicit return.
- C) It returns undefined.
- D) It throws an error.

59. What does "Lexical this" mean in arrow functions?

- A) `this` refers to the global object always.
- B) `this` is inherited from the enclosing scope.
- C) `this` refers to the element that triggered the event.
- D) `this` is null.

60. Which creates a Block Scope?

- A) if statement with `{}`
- B) `function`
- C) `() =>`
- D) Global object

61. What is a Closure?

- A) A function having access to its own scope only.
- B) A function bundled with its lexical environment.
- C) A way to lock a variable.
- D) A function that cannot be exported.

62. Closures are commonly used for:

- A) Data privacy/encapsulation
- B) Speeding up loops
- C) Reducing memory usage
- D) Styling HTML

63. Function Expressions are:

- A) Hoisted
- B) Not hoisted
- C) Always anonymous
- D) Global only

64. Function Declarations are:

- A) Hoisted
- B) Not hoisted

- C) Illegal in strict mode
 - D) Same as variables
65. What output does this produce? `() => { return 5; }()`
- A) undefined
 - B) 5
 - C) Function definition
 - D) Error
66. Can arrow functions be used as constructors (with `new`)?
- A) Yes
 - B) No
 - C) Only with one argument
 - D) Only in classes
67. In a method inside an object, `this` refers to:
- A) The global object
 - B) The object itself (owner)
 - C) The function
 - D) Window
68. Which of the following creates an Immediately Invoked Function Expression (IIFE)?
- A) `function() {}()`
 - B) `(function() {})`
 - C) `function iife() {}`
 - D) `run function() {}`
69. Can you access a variable declared with `var` inside an `if` block from outside that block?
- A) Yes
 - B) No
 - C) Only if it is true
 - D) Only in ES6
70. Can you access a variable declared with `let` inside an `if` block from outside that block?
- A) Yes
 - B) No
 - C) Depends on the browser
 - D) Only if initialized

Part 9: Higher-Order Functions

71. What makes a function "Higher-Order"?
- A) It is declared at the top of the file.
 - B) It takes a function as an argument or returns a function.
 - C) It uses `Math.max`.
 - D) It runs faster.

72. What does `map()` do?

- A) Modifies the array in place.
- B) Creates a new array by applying a function to each element.
- C) Filters elements.
- D) Finds a single element.

73. `filter()` returns:

- A) A single value.
- B) A new array with elements that pass the test.
- C) A boolean.
- D) The first matching element.

74. `reduce()` is best used for:

- A) Iterating without return.
- B) Transforming an array into a new array of same length.
- C) Accumulating an array into a single value (sum, object, etc.).
- D) Sorting.

75. Which method modifies the original array?

- A) `map()`
- B) `filter()`
- C) `sort()`
- D) `reduce()`

76. How do you sort an array of numbers `[10, 2, 5]` correctly?

- A) `arr.sort()`
- B) `arr.sort((a, b) => a - b)`
- C) `arr.sortNumbers()`
- D) `arr.order()`

77. What does `forEach()` return?

- A) The modified array
- B) A new array
- C) `undefined`
- D) The length of the array

78. What does `find()` return?

- A) An array of matches.
- B) The first element that matches the condition.
- C) The index of the match.
- D) Boolean.

79. If `map` is used on an array of length 5, the result array length is:

- A) 0
- B) 1
- C) 5
- D) Variable

80. Does `filter()` modify the original array?

- A) Yes
- B) No

- C) Only if items are removed
 - D) Sometimes
81. Which method checks if *at least one* element passes a test?
- A) `every()`
 - B) `some()`
 - C) `filter()`
 - D) `check()`
82. Which method checks if *all* elements pass a test?
- A) `every()`
 - B) `some()`
 - C) `all()`
 - D) `map()`
83. In `reduce((acc, curr) => ... , init)`, what is `init`?
- A) The final value
 - B) The initial value of the accumulator
 - C) The index
 - D) The array length
84. What happens if you use `sort()` on numbers without a callback?
- A) It sorts numerically.
 - B) It sorts alphabetically (as strings).
 - C) It reverses them.
 - D) It throws an error.
85. Chaining `map().filter()` results in:
- A) A syntax error.
 - B) A single value.
 - C) A processed new array.
 - D) The original array.

Part 10: Mixed Concepts & Syntax

86. How do you write a single line comment in JS?
- A) ``
 - B) `/* */`
 - C) `//`
 - D) `#`
87. What is the correct way to call a function named `myFunction`?
- A) `call myFunction()`
 - B) `myFunction`
 - C) `myFunction()`
 - D) `run.myFunction`
88. Which keyword breaks out of a switch statement?
- A) `stop`
 - B) `break`

C) return

D) exit

89. Which is NOT a logical operator?

A) &&

B) ||

C) !

D) | (Bitwise OR)

90. The do...while loop checks the condition:

A) Before the code block.

B) After the code block.

C) In the middle.

D) Never.

91. Is JavaScript strictly typed?

A) Yes

B) No, it is dynamically typed.

C) Only in classes.

D) Yes, like Java.

92. Which method removes the last element of an array?

A) shift()

B) pop()

C) push()

D) remove()

93. Which method adds an element to the beginning of an array?

A) unshift()

B) push()

C) concat()

D) prepend()

94. Which ES6 feature allows unpacking values from arrays or properties from objects?

A) Spread

B) Rest

C) Destructuring

D) Hoisting

95. `const { name } = person;` is an example of:

A) Object Cloning

B) Object Destructuring

C) Reference Copy

D) Renaming

96. `let arr2 = [...arr1];` is an example of:

A) Rest Operator

B) Spread Operator

C) Concat

D) Loop

97. What is the output of `console.log(typeof NaN)`?
A) NaN
B) undefined
C) number
D) object
98. Which function parses a string to an integer?
A) `Integer.parse()`
B) `parseInt()`
C) `toInteger()`
D) `Number.int()`
99. What is the global object in Node.js?
A) `window`
B) `root`
C) `global`
D) `self`
100. Which feature allows embedding expressions in strings like `${name}`?
A) Concatenation
B) Template Literals
C) String Interpolation (in Python)
D) String Builder

Answer Key

1. A | 2. B | 3. B | 4. C | 5. C
2. C | 7. C | 8. A | 9. C | 10. C
3. C | 12. A | 13. B | 14. B | 15. C
4. C | 17. B | 18. C | 19. C | 20. C
5. C | 22. C | 23. B | 24. B | 25. B
6. B | 27. C | 28. A | 29. B | 30. C
7. B | 32. B | 33. B | 34. A | 35. B
8. C | 37. C | 38. B | 39. B | 40. B
9. C | 42. C | 43. B | 44. C | 45. B
10. A | 47. A | 48. B | 49. B | 50. B
11. B | 52. C | 53. C | 54. C | 55. B
12. B | 57. C | 58. B | 59. B | 60. A
13. B | 62. A | 63. B | 64. A | 65. B
14. B | 67. B | 68. B | 69. A | 70. B
15. B | 72. B | 73. B | 74. C | 75. C
16. B | 77. C | 78. B | 79. C | 80. B
17. B | 82. A | 83. B | 84. B | 85. C
18. C | 87. C | 88. B | 89. D | 90. B
19. B | 92. B | 93. A | 94. C | 95. B

20. B | 97. C | 98. B | 99. C | 100. B